DtP & PtD in DLR

DLR

Problem: density $f(t,x,v)$, $x \in \mathbb{R}^3$, $v \in \mathbb{R}^3$

$$\partial_t f = -v\,\partial_x f + E\,\partial_v f =: RHS(f)$$

Low-rank fluid

$$f(t,x,v) = \sum_{i,j=1}^{r} U_i(t,x) S_{ij}(t) V_j(t,v) \qquad r \ll N$$

DLR : projector

$$\partial_t f + P(f)\,(v\,\partial_x f - E\,\partial_v f) = 0$$

$$P(f)g = P_U g + P_V g - P_U P_V g$$

2 problems : $\begin{cases} \langle U_i V_j, RHS \rangle_{xv} & \text{unfold} \quad O(r^2 N^d) \\ S^{-1} & \end{cases}$

$\Rightarrow$ Stable Integral

PSI : K-S-L . avoid S^{-1}, but Inverse integral of S

BUG $\qquad \dot{Y} = \dot{U} S V^T + U \dot{S} V^T + U S \dot{V}^T$

augmented BUG $\qquad P_Y Z$

PtD or DtP

$$\begin{array}{ccc} \text{PDE} & \xrightarrow{\text{Discretize }(x,v)} & \text{matrix PDE} \quad dF/dt = L_\Delta(F) \\ \downarrow \text{DLR} & & \downarrow \text{DLR Projector} \\ \text{K-S-L} & \longrightarrow & \text{Discretized K-S-L equations} \end{array}$$

Numerical Stability

$$f^{n+1} = G f^n$$

stability : $\|f^n\| \le C \|f^0\|$

e.g. $\partial_t f + a\,\partial_x f = 0$.

中心差分 $\quad 1 - i\nu\sin\theta \quad |g|^2 = 1 + \nu^2 \sin^2\theta > 1 \quad$ unstable. $\qquad f_j^{n+1} = f_j^n - \frac{\nu}{2}(f_{j+1}^n - f_{j-1}^n)$

Upwind $\quad 1 - \nu(1 - e^{-i\theta}) \quad |g| \le 1 \Leftrightarrow 0 \le \nu \le 1 \qquad f_j^{n+1} = f_j^n - \nu(f_j^n - f_{j-1}^n)$

Lax-Friedrichs $\quad \cos\theta - i\nu\sin\theta \quad |g| \le 1 \Leftrightarrow -1 \le \nu \le 1 \qquad f_j^{n+1} = \frac{1}{2}(f_{j+1}^n + f_{j-1}^n) - \frac{\nu}{2}(f_{j+1}^n - f_{j-1}^n)$

Def D : $\quad N = 8 \quad \Delta x = 1/8 \quad (Sf)_j = f_{j-1} \quad (S^{-1}f)_j = f_{j+1} \quad$ S: Moving matrix

$$D_c = \frac{S^{-1} - S}{2\Delta x} \qquad D_{up} = \frac{I - S}{\Delta x} \qquad D_{xx} = \frac{S^{-1} - 2I + S}{\Delta x^2}$$

e.g. D_c :

$$\begin{bmatrix} 0 & 4 & \cdots & 0 & -4 \\ -4 & 0 & 4 & \cdots & 0 \\ 0 & -4 & 0 & 4 \cdots & 0 \\ & & \vdots & & \\ 4 & 0 & 0 & \cdots -4 & 0 \end{bmatrix}$$

$$D_{up} = D_c - \frac{\Delta x}{2} D_{xx}$$

$$RHS_{LF}(f) = -a D_c f + \frac{\Delta x^2}{2\Delta t} D_{xx} f$$

For S (Inverse time)

$$\dot{S} = -U^T F(USV^T) V$$

倒着解热方程不适定

Stability analysis

SAT : $\tilde{F} = F^n + \Delta t\, RHS^{(\Delta)}(F^n)$

$F^{n+1} = T_r(F^n)$

$PtD = DtP$

$\| F^{n+1} \| = \| T_r(\tilde{F}) \| \le \| \tilde{F} \| \le \max_\theta |g(\theta)| \cdot \| F^n \|$

DLR: $\partial_t f = F(f) = -v\partial_x f + E\partial_v f$ linear for f

$\Rightarrow f^{n+1} = G f^n$ G fixed for f

$\partial_t f = P(f) F(f)$ nonlinear for f

$\|G\|$ can't be defined

(This prevents a straightforward approach to analyzing the stability of DLR discretizations)

View of [100]

1. linear in each step.

K : $\dot{K} = F(KV^T)V$ linear for K

2. QR : $K = QR$ $\|K\|_F = \|R\|_F$

SVD : $\| T_r K \|_F \le \|K\|_F$

$\Rightarrow |g_K| \times 1 \times |g_S| \times |g_L| \times (\le 1)$

DtP & PtD

$f = \sum_{i,j} U_i(x) S_{ij} V_j(v)$

$K := US$ $L := VS^T$

DtP : Discretize RHS

$\partial_x U_j (vV_j) \rightsquigarrow (D_x^+ U_j) \otimes (v^+ * V_j) + (D_x^- U_j) \otimes (v^- * V_j)$

$\Rightarrow F \in \mathbb{R}^{N_x \times N_v}$ $F_{ab} \approx f(x_a, v_b)$

$RHS^\Delta(F) = -\sum_\pm (D_x^\pm F)\, \mathrm{diag}(v^\pm) + \sum_\pm \mathrm{diag}(E^\pm)\, F\, (D_v^\pm)^T$

K step : $\partial_t K_i = \langle RHS^\Delta(KV^T), V_i \rangle_v$

S step : $\partial_t S_{ij} = -\langle RHS_S^\Delta, U_i V_j \rangle_{xv}$

$\dot{S} = -U^T M_x RHS^\Delta(USV^T) M_v V$

L step : $\dot{L}^T = U^T M_x RHS^\Delta(UL^T)$

PtD : $\langle U_i V_j, RHS_{LR} \rangle_{xv} = -\sum_{kl} \underbrace{\langle U_i \partial_x U_k \rangle_x}_{C_{ik}^x} S_{kl} \underbrace{\langle V_j v V_l \rangle_v}_{C_{jl}^v}$

K : $\partial_t K_i = -\sum_l C_{il}^v \partial_x K_l$

S : $\partial_t S_{ij} = +\sum_{kl} C_{ik}^x S_{kl} C_{jl}^v$

L : $\partial_t L_i = -v \sum_k C_{ik}^x L_k$

Discretize : $\partial_x K_l \longrightarrow D_K$

$C_{ik}^x = \langle U_i \partial_x U_k \rangle_x \longrightarrow D_S(M_x)$

$C_{jl}^v = \langle V_j v V_l \rangle_v \longrightarrow M_v$.

Augmented BUG

$S_{ij}^{n+1} = S_{ij}^n + \Delta t\, \hat{U}^T M_x RHS^\Delta(f^n) M_v \hat{V}$

$\Rightarrow f_\Delta^{n+1} = \hat{U}\hat{U}^T (f_\Delta^n + \Delta t\, RHS^\Delta(f_\Delta^n)) \hat{V}\hat{V}^T$

$\Rightarrow \| f_\Delta^{n+1} \|_F \le \| f_\Delta^n + \Delta t\, RHS^\Delta(f_\Delta^n) \|_F \longrightarrow$ Stable

Projector Splitting

Reference [100] $\partial_t u(t,x) = -A\,\partial_x u(t,x)$. $u(t,x) \in \mathbb{R}^{N+1}$

Thm 1: DtP ($D_S = D_K$) Unstable.

Thm 3: PtD ($D_S \neq D_K$) $CFL \le 1$

Thm

1	DtP : $D_S = D_K = LF$	$CFL = 3$
2	PtD : 中心差分 . S not stablize	$CFL \le 1/\sqrt{3} = 0.577$
3	PtD : $D_S \neq D_K$. 粘性, change sign	$CFL \le 1$
4	BUG :	$CFL \le 1$